

Larsen Bier

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A signals and controls engineer with a rigorous mathematical background who loves carefully analyzing technical problems. Specializes in system modeling, computer vision, signal processing, and data-driven (and ML) solutions. Always looking to learn and improve.

Education

UNIVERSITY OF CALIFORNIA – LOS ANGELES

M.S., Controls Engineering (Mechanical & Aerospace Engineering)
B.S., Mathematics, Minor in Data Science Engineering

Expected 2026-2028
2022-2025

GPA: N/A
GPA: 3.981

Work Experience

CONTROLS ENGINEER INTERN, ASML

Jun 2026 — Present, San Diego, CA

- Researched, implemented, and designed tests for a high speed optical measurement algorithm for bubbles at small scales.
- Translated algorithm requirements into test-bench lighting and framerate specs, cutting signal noise and improving measurements.

COMPUTER VISION RESEARCHER, EPI Imaging

Jun 2025 — Sep 2025, Palo Alto, CA

- Developed statistical algorithms to batch-test 3D camera calibrations, enabling scalable detection of miscalibrated units.
- Trained a ResNet classifier to distinguish empty from connected cable ports, achieving >95% accuracy.
- Ran pose-estimation experiments with 5 evaluation metrics to confirm and correct a systematic 5% pose error in the optics system.
- Automated translating camera specifications into Blender camera simulations, saving up to 30 minutes per experiment.

DATA SCIENCE INTERN, Chipotle Mexican Grill

Jun 2023 — Aug 2023, Newport Beach, CA

- Identified and quantified region-specific increases in delivery sales during adverse weather via multivariate regression, delivering actionable effect estimates to marketing for policy use.
- Coded a scalable SQL pipeline to query millions of rows across 500+ stores for marketing analytics.

Research & Academic Experience

RESEARCH ASSISTANT, UCLA GLACIER Lab (Dr. Shahriar Talebi)

Jan 2025 — Present, Los Angeles, CA

- Designed 5 test protocols to validate novel data-driven learning algorithms, documenting each thoroughly to ensure reproducibility.
- Led experiments uncovering empirical hyperparameter ranges neglected by theory, enabling effective use of optimization algorithms.

PROJECT MANAGER, UCLA DataRes Club

Jan 2024 — Jun 2025, Los Angeles, CA

- Directed technical strategy for a 10-week Connect4 AI software project, guiding the team towards a Monte-Carlo tree search approach, resulting in a master-level AI agent.

Publications

Bier, Larsen and Talebi, Shahriar, “[Learning Kalman Policy for Singular Unknown Noise Covariances via Riemannian Regularization.](#)” IEEE Conference on Decision and Control (CDC), 2026. (accepted, pending minor revisions).

Relevant Skills

[Python](#) [MATLAB](#) [C++](#) [SQL](#) [GitHub](#) [LaTeX](#) [PID Control](#) [LQR Control](#) [Data Analysis](#) [Machine Learning](#) [Computer Vision](#)

[Robot Operating System \(ROS\)](#) [Statistics](#) [Optimization Algorithms](#) [MS Office Suite](#) [Spanish \(Fluent\)](#) [AI Agent Workflows](#)